FIELD GUIDE · VOL. 01

TRACK WITH TECH

let's just find one more signal

A developer's handbook to tracking, inspired by **Tracker**: how location technology really works, how to find your own things, how to **build** a tracker with Android + Supabase + Next.js, how to track software and live data, and how to train real search skills through **ethical OSINT**.

> **Rule zero:** track **things you own**, **data that is public**, or **people who said yes**. Covert tracking of a person is stalking. It is a crime in India under the Bharatiya Nyaya Sanhita, and in most other countries.

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THE TRACKER MINDSET

Colter Shaw in *Tracker* does not guess. He **reads signals**, makes **hypotheses**, puts **odds** on each one, and follows the most likely trail first. That is the same loop as debugging, analytics and OSINT. Tech only gives you better signals.

▲ 1 · SIGNAL

What data exists? GPS fix, Wi-Fi scan, log line, photo metadata, a timestamp.

▲ 2 · HYPOTHESIS

Where could it be? List options and give each one a %. "70% left it at college, 20% in the bus..."

▲ 3 · VERIFY

Find a second, independent signal. One source is a clue, and two sources are a fix.

FIELD MAP: WHAT'S INSIDE

#	SECTION	YOU WILL BE ABLE TO
02	How location tech works	Explain GPS, Wi-Fi, cell, BLE, UWB and IP, and how accurate each one is
03	Track your own things	Set up phone, laptop, bag, bike and pet tracking properly
04	Build a tracker	Android app → Supabase/PostGIS → live map, plus geofences
05	Hardware tracker	ESP32 + GPS module + SIM/LoRa that you own and control
06	Track your software	Analytics, crashes, uptime and builds for apps like Stack
07	Track the live world	Planes, ships, satellites, parcels, prices and web pages through APIs
08	OSINT: think like a tracker	Geolocate photos and use public data the right way
09	Rules & counter-tracking	Know the law; detect if **you** are being tracked
10	30-day roadmap	Build 5 portfolio projects

02

HOW LOCATION TECH WORKS

Every "where is it?" answer comes from one of these signals. Knowing their **accuracy** is the most important tracker skill, because it tells you how big your search area is.

SIGNAL	HOW IT WORKS	TYPICAL ACCURACY	NEEDS
GNSS / GPS	Time signals from 4+ satellites (GPS, GLONASS, Galileo, India's NavIC) → trilateration	~3–10 m outdoors; poor indoors	Sky view, power
A-GPS	The network sends satellite data so the first fix takes seconds, not minutes	Same as GPS, faster	Data connection
Wi-Fi positioning	Nearby router IDs (BSSID) are matched against a database of known locations	~10–40 m; works indoors	Wi-Fi on
Cell towers	Which towers you hear and how strongly (cell ID / timing)	~100 m in cities to several km in rural areas	SIM / radio
Bluetooth LE	A beacon broadcasts an ID; nearby phones report seeing it (crowd network)	Proximity, ~1–30 m	Phones nearby
UWB	Ultra-wideband time-of-flight gives distance and direction	~10–30 cm	UWB chips both ends
IP geolocation	The IP address is looked up in a registry or commercial database	Country to city, often wrong	Internet
Photo EXIF	The camera writes GPS coordinates into image metadata	As good as the phone's GPS	Location tags on

FUSED LOCATION = COMBINE SIGNALS

Android's `FusedLocationProviderClient` blends GPS, Wi-Fi, cell and motion sensors, and you choose the trade-off:

`HIGH_ACCURACY` GPS-first, ~5 m, heavy on battery. Use it for live navigation.

`BALANCED_POWER` Wi-Fi/cell, ~40–100 m, light on battery. Use it for "which city / campus".

Tracker tip: every fix comes with an `accuracy` radius. Always store it, and draw it as a circle on the map, not a pin. A pin with a 2 km radius is a lie.

03

TRACK YOUR OWN THINGS

Before you build anything, set these up. They are free, legal, and handle 90% of "I lost it" cases.

THING	TOOL	SETUP CHECKLIST
Android phone	Google **Find My Device** (crowd network)	Settings → Google → Find My Device → ON; enable "store recent location" and "find offline"; learn the website so you can ring, lock or erase the phone
Bag / keys / wallet	Bluetooth tags (Find My Device-compatible tags, Samsung SmartTag, Tile, AirTag for iPhone users)	Pick a tag that matches **your** phone's network; label it; check battery monthly
Laptop (Linux)	Prey (open source tier) or a self-hosted script that posts IP + Wi-Fi scan	Set a BIOS password and full-disk encryption; note the serial and MAC address
Bike / scooter	GPS tracker with a SIM (4G) or a hidden BLE tag	Hidden mount, motion alerts, geofence around home
Pet	GPS collar or BLE tag	Geofence alerts when it leaves home
Family (consent)	Google Maps location sharing, Life360	Mutual, visible, and either person can turn it off at any time

WHEN SOMETHING IS LOST: THE TRACKER DRILL

FIRST 10 MINUTES

- Open Find My Device → last-seen time and the accuracy circle
- Ring it if it's within range
- Check Google Maps Timeline (if enabled) for the path

IF STOLEN

- Lock the phone and show a message; **do not** confront anyone yourself
- Note the **IMEI** (dial *#06#) → file a police report → block it on India's **CEIR** portal (sancharsaathi.gov.in)
- Change passwords from another device

BUILD A TRACKER: YOUR STACK

You already know Android (Capacitor), Supabase and Next.js. That is a complete tracking pipeline: **phone → database → live map**. Build it for **your own devices** or for friends who opt in.

```
[ Android app ] ──GPS fix──▶ [ Supabase · PostGIS table ] ──realtime──▶ [ Next.js + MapLibre map ]
                                                          ──▶ geofence alert
```

1 · DATABASE (SUPABASE SQL)

```sql
create extension if not exists postgis;
create table pings (
  id bigserial primary key,
  device_id uuid references devices(id),
  owner uuid default auth.uid(),          -- only the owner can read
  geom geography(Point, 4326) not null,
  accuracy_m real, battery int,
  created_at timestamptz default now()
);
alter table pings enable row level security;
create policy "own pings" on pings for all using (owner = auth.uid());
```

2 · ANDROID: SEND A FIX (KOTLIN)

```kotlin
val client = LocationServices.getFusedLocationProviderClient(ctx)
client.getCurrentLocation(Priority.PRIORITY_BALANCED_POWER_ACCURACY, null)
  .addOnSuccessListener { loc ->
    supabase.from("pings").insert(mapOf(
      "device_id" to deviceId,
      "geom" to "POINT(${loc.longitude} ${loc.latitude})",
      "accuracy_m" to loc.accuracy))
  }
// Background = foreground service + visible notification ("Tracking is ON")
```

3 · LIVE MAP (NEXT.JS)

```ts
supabase.channel("pings")
  .on("postgres_changes", { event: "INSERT", schema: "public", table: "pings" },
      ({ new: p }) => marker.setLngLat(toLngLat(p.geom)))   // MapLibre / Leaflet
  .subscribe();
```

4 · GEOFENCE ALERT (POSTGIS)

```sql
select ST_DWithin(p.geom, z.center, z.radius_m) as inside
from pings p, zones z where p.id = $1;   -- inside → outside? send a push
```

Design for honesty: a persistent notification, a big OFF switch, auto-delete pings after 30 days, and no hidden mode. That is also what Google Play policy requires for background location.

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HARDWARE TRACKER (FOR YOUR BIKE / BAG)

A ₹1,500–2,500 DIY tracker teaches embedded systems, radio and power management, which makes it a strong résumé project.

PART	EXAMPLE	JOB
Microcontroller	ESP32 (Wi-Fi + BLE)	The brain; can also do Wi-Fi scan positioning
GNSS module	u-blox NEO-6M / NEO-M8N	Outputs NMEA sentences with lat/lon
Uplink	SIM800L / SIM7600 (2G/4G) *or* LoRa SX1276	Sends the fix to your server (LoRa = long range, no SIM)
Power	18650 Li-ion + TP4056 charger	Deep sleep between fixes → weeks of battery
Motion	MPU6050 accelerometer	Wakes the board only when the bike moves

FIRMWARE SKETCH (ARDUINO / ESP32)

```
#include <TinyGPSPlus.h>
TinyGPSPlus gps;  HardwareSerial GPS(1);
void setup(){ GPS.begin(9600, SERIAL_8N1, 16, 17); connectWiFiOrModem(); }
void loop(){
  while (GPS.available()) gps.encode(GPS.read());
  if (gps.location.isUpdated() && gps.hdop.hdop() < 2.5) {     // good fix only
    postJSON("/rest/v1/pings", gps.location.lat(), gps.location.lng());
    esp_sleep_enable_timer_wakeup(5 * 60 * 1000000ULL);           // 5 min
    esp_deep_sleep_start();
  }
}
```

READ THE RAW SIGNAL: NMEA

```
$GPGGA,123519,2438.82,N,07718.68,E,1,08,0.9,545.4,M,...
       time    lat         lon        fix sats HDOP altitude
```

LEVEL UP

- Send to The Things Network (LoRaWAN), which is free
- Kalman filter to smooth jumpy fixes
- Battery % in every ping

ONLY ON YOUR STUFF

Putting a tracker on someone else's vehicle or bag without consent is illegal. Build it for your own bike, bag, or a friend's that they set up with you.

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TRACK YOUR SOFTWARE

In the industry, "tracking" mostly means **knowing what your app is doing**. This is the most hireable tracking skill, and you can apply it to Stack today.

TRACK	TOOL (FREE TIER)	QUESTION IT ANSWERS
Product analytics	PostHog, Umami, Plausible	Which tabs do people use? Where do they drop off?
Crashes / errors	Sentry, Firebase Crashlytics	What broke, on which phone, after which release?
Uptime	UptimeRobot, Better Stack	Is my API / Supabase down right now?
Logs	Supabase logs, Axiom, Grafana Loki	What exactly happened at 12:38?
Builds & releases	GitHub Actions, Play Console	Which commit is live? Did `npm run build` pass?
Code time	WakaTime	How many hours did Stack really take?

A CLEAN EVENT SCHEMA (STACK EXAMPLE)

```
track("pdf_opened",     { source: "telegram", pages: 312 })
track("highlight_made",  { doc_type: "pdf" })
track("focus_session",   { minutes: 25, music: "local" })
// object_verb names · no titles, no note text, no personal data
```

GOOD TRACKING

- Anonymous IDs
- An opt-out switch in Settings
- Disclosed in the privacy policy (Stack has `/privacy`)
- Data deleted with the account

FUNNEL TO WATCH

- Install → first PDF added
- First PDF → first highlight
- Day 1 → day 7 return
- That last step shows whether the Gen Z UI work paid off

India's **DPDP Act 2023** requires clear consent and a stated purpose for personal data, so design for it from day one.

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TRACK THE LIVE WORLD

Huge amounts of tracking data are **public by design**. They make great practice and great side projects.

TARGET	PUBLIC SIGNAL	WHERE TO GET IT	PROJECT IDEA
Planes	ADS-B (aircraft broadcast position)	OpenSky Network API; or your own RTL-SDR dongle + dump1090	"Flights over Guna right now" map
Ships	AIS transponders	AISHub, MarineTraffic	Port traffic dashboard
Satellites / ISS	Orbital elements (TLE)	CelesTrak, N2YO API, satellite.js	"ISS visible tonight" alert
Trains	Running status	NTES / railway status APIs	Late-train notifier
Parcels	Courier tracking IDs	AfterShip, 17TRACK APIs	One page for all your orders
Prices	Product pages	changedetection.io (self-host), Keepa	Price-drop Telegram bot
Web pages	Page diffs, RSS	changedetection.io, RSS feeds, Wayback Machine	Exam-result / job-post watcher
Code	Repos, releases	GitHub API / watch, npm registry	Dependency release tracker

PATTERN YOU WILL REUSE EVERYWHERE

```
// poll → diff → store → alert   (Vercel cron / Supabase Edge Function)
const now  = await fetchSource();
const prev = await db.latest(key);
if (changed(prev, now)) { await db.insert(now); await notify(telegram, diff(prev, now)); }
```

You already run a Telegram bot for Stack imports. Point the same bot at a tracker and you have a **personal alerts system**.

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OSINT: THINK LIKE A TRACKER

Open-source intelligence means finding answers using only **public** information. Journalists (Bellingcat), researchers and missing-person volunteers use it. It is the closest real skill to what Colter Shaw does.

PHOTO GEOLOCATION: "WHERE WAS THIS TAKEN?"

READ THE IMAGE

- Signs, language, number plates, shop names
- Road markings, poles, plants, terrain
- Shadows → direction and time (SunCalc)
- Landmarks on the skyline

THEN VERIFY

- Reverse image search: Google Lens, Yandex, TinEye
- Match with Google Street View / Earth 3D
- Query map features: Overpass Turbo ("church next to a river and a petrol pump")
- EXIF: `exiftool photo.jpg` (often stripped by social apps)

PRACTICE LEGALLY: THIS IS WHERE THE SKILL COMES FROM

PLATFORM	WHAT YOU DO
GeoGuessr / OpenGuessr	Guess locations from Street View; trains your eye fast
Bellingcat guides & challenges	Real investigation techniques and open tool lists
Trace Labs Search Party CTF	Teams use OSINT to find leads on **real missing persons** cases and pass them to law enforcement. It is literally "Tracker" for good
OSINT Framework (osintframework.com)	A map of public tools by category
Quiztime (community challenges)	Daily geolocation puzzles

Ethics line: OSINT is for public questions (events, places, missing-persons work, verifying fake news). Don't use it to profile, expose or follow a private person. Doxxing and stalking cause real harm and are illegal.

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RULES & COUNTER-TRACKING

GREEN ZONE

- Your own devices and things
- Opt-in family/friend sharing that both people can see
- Your app's analytics, with consent and a privacy policy
- Public data: flights, ships, prices, public posts about events
- Authorized CTFs and missing-person OSINT events

RED ZONE

- Hidden trackers on someone's car, bag or phone
- Spyware or "parental" apps installed secretly on an adult's phone
- Tracking a partner, ex or classmate
- Using leaked databases or someone's account
- Sharing a private person's location or address

In India this can fall under **stalking** (Bharatiya Nyaya Sanhita 2023), the **IT Act 2000** (unauthorised access, privacy violation) and the **DPDP Act 2023**. Other countries have similar laws.

AM I BEING TRACKED? (PROTECT YOURSELF AND FRIENDS)

CHECK	HOW
Unknown Bluetooth trackers	Android: Settings → Safety & emergency → **Unknown tracker alerts** → Scan now
Location sharing	Google Maps → profile → Location sharing; remove anyone unexpected
Apps with location access	Settings → Location → App location permissions → look for "Allowed all the time"
Device admin / accessibility abuse	Settings → Security → Device admin apps; Accessibility → unknown services (common spyware tricks)
Logged-in sessions	Google account → Security → Your devices; sign out of unknown ones
Photo metadata	Strip location before sharing (Google Photos → share settings / Scrambled Exif app)

The best trackers are also the best at **not being tracked**. Run this check on your own phone tonight.

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30-DAY ROADMAP

WEEK	LEARN	BUILD (PORTFOLIO)
1 Signals	GPS/Wi-Fi/cell accuracy, NMEA, EXIF, Find My Device setup, counter-tracking checklist	**Project 1:** "Where was this photo?" web tool. Upload a photo → read EXIF → show it on a map with an accuracy circle (Next.js + exifr + MapLibre)
2 Pipeline	Fused location, foreground services, Supabase PostGIS, realtime, RLS	**Project 2:** "My Devices" tracker. Your phone + laptop ping Supabase; live map, history trail, geofence push
3 Live data	REST polling, cron, diffing, Telegram bots, OpenSky / CelesTrak APIs	**Project 3:** Sky Tracker, with planes + ISS over your city. **Project 4:** page/price watcher bot
4 OSINT + HW	Geolocation method, Overpass Turbo, SunCalc; ESP32 + GPS basics	**Project 5:** ESP32 bike tracker → same Supabase backend. Join a Trace Labs CTF or 20 Quiztime puzzles

SKILL CHECKLIST

- ▲ **Can explain** why a pin has an accuracy radius
- ▲ **Can build** phone → PostGIS → live map
- ▲ **Can geolocate** a street photo using only public tools
- ▲ **Can instrument** Stack with privacy-safe analytics
- ▲ **Can detect** unknown trackers and stalkerware
- ▲ **Knows the line:** consent, ownership, public data

FIND THE SIGNAL.
RESPECT THE PERSON.

npm run build, then let's just make one more change.